

Solutions to Exercises of Mathematical Preliminaries on Metric Spaces

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Exercise 1. Let S be the collection of continuous functions on $[0, 1]$. Check if

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)|$$

is a valid metric on S .

(Solution) We check each property of a metric.

(1) For identity, we check both sides.

- ($\Rightarrow$) Suppose $d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)| = 0$. Then for every $x \in [0, 1]$, we have

$$0 \leq |f(x) - g(x)| \leq d(f, g) = 0.$$

Hence, $|f(x) - g(x)| = 0$, so $f(x) = g(x)$ for all $x \in [0, 1]$.

- ($\Leftarrow$) Suppose $f(x) = g(x) \forall x \in [0, 1]$. Then it naturally holds that

$$d(f, g) = \sup_{x \in [0, 1]} |f(x) - g(x)| = 0$$

Therefore, $d(f, g) = 0 \iff f = g$.

(2) We can show symmetry by

$$\begin{aligned} d(f, g) &= \sup_{x \in [0, 1]} |f(x) - g(x)| \\ &= \sup_{x \in [0, 1]} |g(x) - f(x)| \cdots \text{by the absolute value} \\ &= d(g, f) \end{aligned}$$

*Comments and corrections are welcome and may be sent to hchu@nd.edu.

(3) We can show triangle inequality by

$$\begin{aligned}
d(f, g) &= \sup_{x \in [0,1]} |f(x) - g(x)| = \sup_{x \in [0,1]} |f(x) - h(x) + h(x) - g(x)| \\
&\leq \sup_{x \in [0,1]} \{|f(x) - h(x)| + |h(x) - g(x)|\} \\
&= \sup_{x \in [0,1]} |f(x) - h(x)| + \sup_{x \in [0,1]} |h(x) - g(x)| \cdots \text{by triangle inequality} \\
&= d(f, h) + d(h, g)
\end{aligned}$$

□

Exercise 2. Prove or disprove each of the following statements.

(1) An intersection of infinite open sets is open.

(2) An union of infinitely many closed sets is closed.

(Solution) Each statement can be disproved using the counterexamples as follows.

(1) Suppose $S_n = (-\frac{1}{n}, \frac{1}{n})$, which is an open set. Then $\cap_{n=1}^{\infty} S_n = \{0\}$, which is not open as it is a single point.

(2) Suppose $S_n = [-1 + \frac{1}{n}, 1 - \frac{1}{n}]$, which is closed. Then $\cup_{n=1}^{\infty} S_n = (-1, 1)$, which is not closed and is open.

□

Exercise 3. Show that if $x_n \leq y_n \leq z_n$ for all $n \in \mathbb{N}$, and if $\lim x_n = \lim z_n = l$, then $\lim y_n = l$ as well.

(Solution) Take an arbitrary $\varepsilon > 0$ and let N be sufficiently large such that

$$|x_n - l| < \frac{\varepsilon}{4} \quad \text{and} \quad |z_n - l| < \frac{\varepsilon}{2}.$$

Then we have

$$|x_n - z_n| = |x_n - l - z_n + l| \leq |x_n - l| + |z_n - l| < \frac{\varepsilon}{2}$$

by triangle inequality. Also, since we have $x_n \leq y_n \leq z_n$,

$$|y_n - x_n| = y_n - x_n \leq z_n - x_n = |z_n - x_n|.$$

Thus,

$$\begin{aligned}
|y_n - l| &= |y_n - x_n + x_n - l| \\
&\leq |y_n - x_n| + |x_n - l| \cdots \text{by triangle inequality}
\end{aligned}$$

$$\begin{aligned} &\leq |z_n - x_n| + |x_n - l| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{4} < \varepsilon. \end{aligned}$$

Therefore, $\lim y_n = l$. □

Exercise 4. Let (a_n) and (b_n) be Cauchy sequences. Decide whether each of the following sequences is a Cauchy sequence, justifying each conclusion.

(1) $c_n = |a_n - b_n|$

(2) $c_n = (-1)^n a_n$

(Solution) (1) Note that

$$\begin{aligned} c_n &= |a_n - b_n| \\ &= |a_n - a_m + b_m - b_n + a_m - b_m| \\ &\leq |a_n - a_m| + |b_m - b_n| + |a_m - b_m| \end{aligned}$$

by triangle inequality. The above can be written as

$$\Rightarrow |a_n - b_n| - |a_m - b_m| \leq |a_n - a_m| + |b_n - b_m| \quad (1)$$

and

$$\Rightarrow |a_m - b_m| - |a_n - b_n| \leq |a_n - a_m| + |b_n - b_m| \quad (2)$$

Thus, combining Equation (1) and (2), we get

$$||a_n - b_n| - |a_m - b_m|| \leq |a_n - a_m| + |b_n - b_m|$$

Since a_n and b_n are both Cauchy, $\exists N_1$ and N_2 such that

$$|a_n - a_m| < \frac{\varepsilon}{2} \quad \forall n, m \geq N_1 \quad \text{and} \quad |b_n - b_m| < \frac{\varepsilon}{2} \quad \forall n, m \geq N_2.$$

Take $N = \max\{N_1, N_2\}$. Then we have

$$||a_n - b_n| - |a_m - b_m|| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

which implies that $c_n = |a_n - b_n|$ is Cauchy.

(2) Consider $a_n = 1$ for all $n \in \mathbb{N}$. Then a_n is Cauchy but

$$c_n = (-1)^n a_n = (-1)^n$$

will not be Cauchy as it alternates between -1 and 1 . □